ITIL 4

Information Technology Infrastructure Library

Contents

ITIL Service Definition 3

User 3

Customer 4

Sponsor 4

Services 4

Service Offering 4

Product 4

Critical Success Factor (CSF) 4

Key Performance Indicators (KPI) 4

Cost 4

Organisation 4

Output 4

Outcome 5

Service Provider 5

Service Consumer 5

Value Co-creation 5

Utility & Warranty 6

Differences between traditional Project Management vs ITIL 6

ITIL V4 Roadmap 7

Importance of Data 7

Responsibility Assignment Matrix (RACI) 8

Benefits 8

The 7 Guiding Principles 10

Focus on Value 10

Start Where You Are 10

Progress Iteratively with Feedback 10

Progress Iteratively with Feedback 11

Collaborate & Promote Visibility 11

Think & Work Holistically 11

Keep it Simple & Practical 11

Optimise & Automate 11
Four Dimensions of Service Management 12
Organisation & People 12
Information & Technology 12
Partners & Suppliers 12
Value Stream & Process 12
ITIL Service Value System 13
Framework of the Service Value System 13
Service Value Chain 14
17 Practices of Service Management 14
14 Practices of General Management 16
3 Practices of Technical Management 17
Six Activities of the Service Value Chain 17
Interconnected Nature of the Six Values 18
Different Types of Value Streams 18
Purpose of Each Service Value Chain 19
Purpose of 15 ITIL 4 Practices 22
Information Security Management 22
Relationship Management 22
Supplier Management 23
Asset Management 23
Release Management 24
Monitoring & Event Management 24
Service Configuration Management 25
Deployment Management 25
Continual Improvement 26
Change Enablement (Change Control) 26
Incident Management 27
Problem Management 27
Service Request Management 28
Service Desk 28
Service Level Management 28

ITIL Service Definition

User

A person who uses the services.

Customer

A person who **defines the requirements** for a service and **takes responsibility for the outcomes** of the service consumption.

Sponsor

A person who **authorises the budget** for a service consumption.

Services

A means of enabling value co-creation by **facilitating outcomes** customers want to achieve without the customer having to manage specific costs & risks.

Service Offering

A package or one or more services designed to address needs that may include goods, access to resources & service actions.

Product

A configuration of an organisation's resources designed to **offer value for a customer.**

Critical Success Factor (CSF)

A necessary precondition for the achievement of **intended results**.

Key Performance Indicators (KPI)

A **metric** used to evaluate success in the **meeting of an objective**.

Cost

Money spent or sacrificed by the supplier or consumer.

Organisation

A person, group or entity with its own functions & objectives supplying or consuming services.

Output

A tangible deliverable produced by an activity, such as a report, a software update, or a delivered service.

- Incident management tool
- Monthly IT performance reports
- Delivered server hardware

Outcome

The result or impact that the output enables, such as improved efficiency, increased customer satisfaction, or reduced downtime.

- Faster resolution and fewer escalations
- Improved decision making by IT leadership
- Increased system availability for business apps

Service Provider

- Seen as the **IT department**.
- Could be **selling services on the open market**:
  - Other businesses or individual customers.
  - Part of a service alliance, collaborating to provide services to consumer organisations.
  - Understanding of who its consumers are.
  - Understands who the other stakeholders are.

Service Consumer

- Involved in **service consumption**, such as customers, users & sponsors.
- **Example:** If a company wants to purchase mobile phones services for its employees from a wireless carrier (service provider), **various consumer roles may be distributed.**
- Each role may **have different & conflicting expectations** of services & definitions of value.

Stakeholders	*Value for Stakeholders*
Service Consumers	Benefits achieved; costs & risks optimised
Service Provider Employees	Financial & non-financial incentives
Society & Community	Employment; taxes, social contributions
Charity Organisations	Financial & non-financial contributions from other organisations
Shareholders	Financial benefits from dividends; sense of assurance & stability

Value Co-creation

- Value is co-created **only** when the service provider and consumer work together.
- This directly ties to why things like **Focus on Value**, **Collaborate** & **Promote Visibility** principles exist

Value = (Benefits + Outcome Achievement + Risk Reduction) - Cost

Utility & Warranty

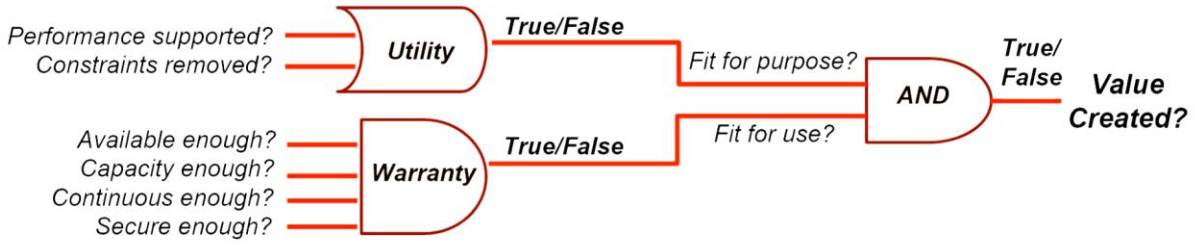
Objective should always be delivering business outcomes & achieving business objectives.

Focus on:

- Utility (fit for purpose)
- Warranty (fit for use)

Both work together to deliver desired outcomes for customers to assess value

- Utility is what the service does
- Warranty is how its delivered



Differences between traditional Project Management vs ITIL

Aspects	*Project Management & Methodologies*	*ITIL & IT Service Management*
Definition	Structured discipline	Best practice framework
Purpose	Delivering on time, in-scope & on-budget	Aligning IT services with business needs
Framework Type	Process based (PMBOK, PRINCE2, Agile, Scrum)	Framework based
Focus	Managing temporary, unique initiatives	Ongoing service delivery
Processes vs. Practices	Process oriented	Practice oriented
Guidelines	Provides detailed guidelines	Provides structured best practices
Approach	Project centric	Service centric
Roles	PM, Sponsor, Stakeholders, Team members	Service Owner, Service Desk, Change Manager, Problem Manager
Lifecycle vs. Value Chain	Project Lifecycle	Service Value Chain
Outcome Measurement	Success measured by customer satisfaction	Success measured by service performance & delivery

ITIL V4 Roadmap

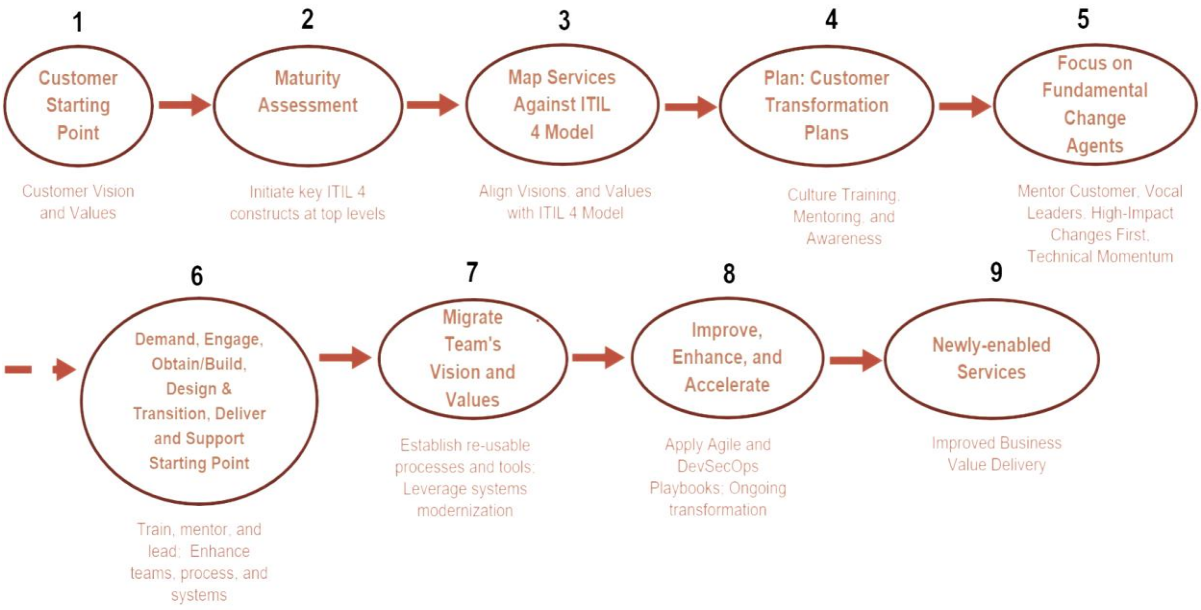
Includes a structured approach to **systematically cover all aspects**.

Aligns the ITIL **implementation** with the organisation's **vision**, values & business goals.

Continuous Improvement (CI) to help the organisation **adapt**, **improve** & **evolve** its **IT service value**.

Ensure Stakeholder Engagement **buy-in** & **support** for this transformation.

Mitigates Risk by **proactively** addressing potential risks and challenges.

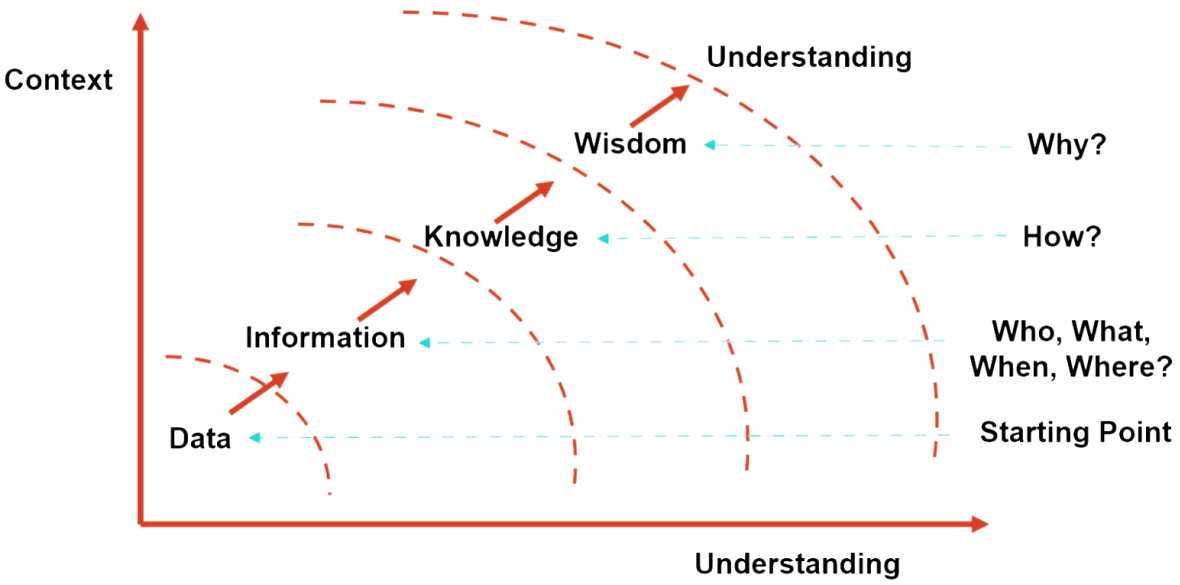


Importance of Data

Log all data pertaining to internal services & maintain the integrity of said data.

Provide customers and users with meaningful content and communication:

- Distinguish Hardware or Software failures



Responsibility Assignment Matrix (RACI)

Individuals may perform multiple tasks across several processes

Two advantages using a RACI matrix:

- Responsibilities for an individual or team
- Workflow & handoff points

CI will use this information to define communication requirements

ACTIVITY	DIRECTOR OF SERVICE MANGEMENT	SERVICE LEVEL MANAGER	PROBLEM MANAGER	SECURITY MANAGER	PROCUREMENT MANAGER
ACTIVITY 1	A, R	C	I	I	C
ACTIVITY 2	A	R	C	C	C
ACTIVITY 3	I	A	R	I	C
ACTIVITY 4	I	A	R	I	-
ACTIVITY 5	I	A	R	C	I

Benefits

Improved Service Management:

- Improved Service Management via comprehensive framework
- Helps organisations implement best practices for service delivery

Better Risk Management:

- Practices to ID, assess & manage risks
- Proactive risk management approach

Enhanced Customer Satisfaction:

- Deliver value to customers
- Customer centric approach

Enhanced Collaboration:

- Encourages collaboration & communication
- Ensures everyone is working towards common goals

Continuous Improvement:

- Emphasises importance of continuous improvement
- Focuses on continuous improvement

Increased Efficiency & Productivity:

- Promotes standard processes & practices
- Helps streamline operations

The 7 Guiding Principles

Emphasise value creation via a comprehensive framework ensuring a balanced & integrated approach.

Focus on Value

Nature: Emphasises everything the organisation does should create value for stakeholders.

Use: Identifies who the stakeholders are and understand their needs & expectations.

Interaction: Interacts with others by ensuring all actions & decisions are aligned with delivering value.

Start Where You Are

Nature: Advises organisations to assess their current state before making changes.

Use: Conducts a thorough assessment of existing processes, services & capabilities.

Interaction: Works with other principles by providing a realistic starting point to improve from.

Progress Iteratively with Feedback

Nature: Encourages making small, incremental changes vs large, disruptive ones.

Use: Implements changes in manageable chunks & seek feedback at each stage.

Interaction: Complements the principle of "Focus on Value" by ensuring each iteration delivers measurable value.

Progress Iteratively with Feedback

Nature: Encourages making small, incremental changes vs large, disruptive ones.

Use: Implements changes in manageable chunks & seek feedback at each stage.

Interaction: Complements the principle of "Focus on Value" by ensuring each iteration delivers measurable value.

Collaborate & Promote Visibility

Nature: Highlights the importance of collaboration & transparency.

Use: Fosters a culture of open communication & teamwork.

Interaction: Supports other principles by ensuring all stakeholders are informed & involved in the process.

Think & Work Holistically

Definition of Holistic: To consider the whole of something as opposed to a part.

Nature: Stresses need to consider the entire organisation & its interdependencies.

Use: Takes a systems thinking approach to understand how different parts of the organisation interact.

Interaction: Ensures changes made in one area do not negatively impact other areas.

Keep it Simple & Practical

Nature: Advises against overcomplicating processes & solutions.

Use: Strives for simplicity & practicality in all actions & decisions.

Interaction: Works with other principles by ensuring solutions are easy to understand & implement.

Optimise & Automate

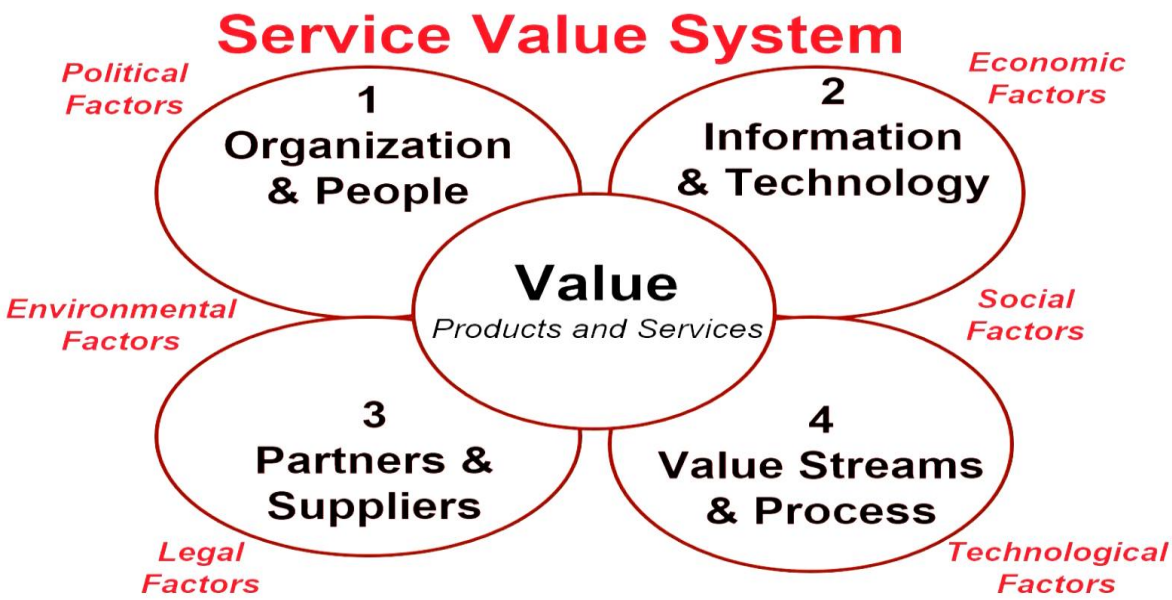
Nature: Focuses on improving efficiency through optimisation & automation.

Use: Identifies opportunities to streamline processes & leverage technology.

Interaction: Supports the principle of "Progress Iteratively with Feedback" by enabling faster & more efficient iterations.

Four Dimensions of Service Management

No one area can be successful in a vacuum, balance between all four areas is necessary to achieve successful service management.



Organisation & People

- Focuses on the human aspect.
- Emphasises importance or organisation culture, structure & roles.
- Focuses on shared values, beliefs & norms.
- Hierarchy & arrangement or roles.
- Clearly defined roles & responsibilities.

Information & Technology

- Ensuring accurate & relevant information.
- Hardware & software used to deliver & support services.
- Protecting information & technology assets.

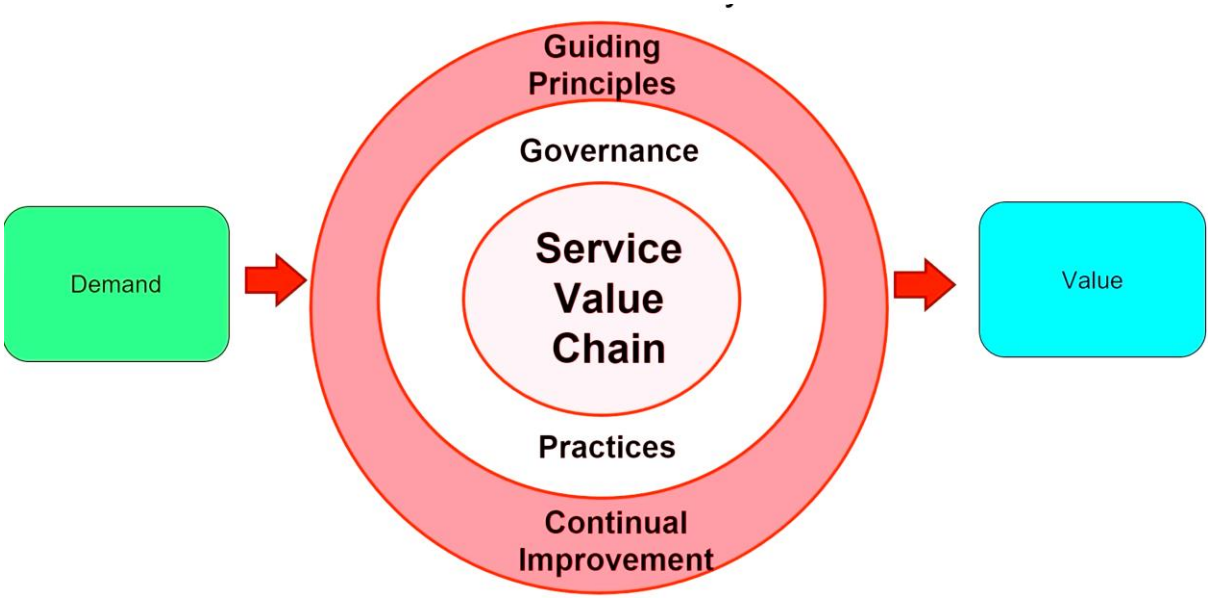
Partners & Suppliers

- Selecting & managing suppliers.
- Building & maintaining relationships with partners.
- Negotiating & managing contracts.

Value Stream & Process

- Series of steps to deliver value to customers.
- Designing, implementing & optimising processes.
- Reviewing & improving processes.

ITIL Service Value System



Framework of the Service Value System

This system shows how all components & activities of the organisation work together to create value. CI is key & at the heart of the system.

ITIL Service Value Chain	ITIL Practices	ITIL Guiding Principles	Governance	Continual Improvement
Plan	General Management Practices	Focus on Value	Directs & Controls Organisation	Seven Step Improvement
Engage	Service Management Practices	Start Where You Are		
Design & transition	Technology Management Practices	Progress Iteratively with Feedback		
Obtain / build		Collaborate & Promote Visibility		
Deliver & support		Think & Work Holistically		
Improve		Keep it Simple & Practical		
		Optimise & Automate		

Service Value Chain

Highlights how value streams utilise different paths through the Service Value Chain.

Dynamic model enables organisations to align IT services with business outcomes.

Highly interconnected.

Service Value Chain activity builds upon and supports a seamless flow of service management.

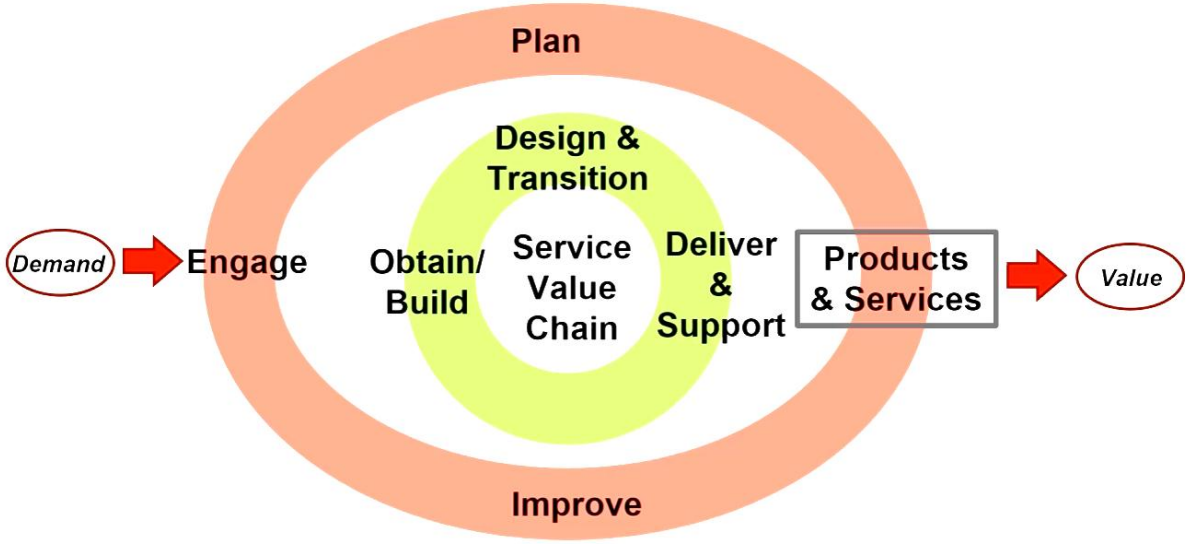
Supports multiple value streams.

Promotes agility, continuous improvement & customer-centric service management.

ITIL 4 Practices integrate into the Service Value Chain.

Value Streams create specific workflows that leverage different parts of the Service Value Chain.

Organisations can tailor the value streams based different service needs.



17 Practices of Service Management

Availability Management

Purpose: to ensure agreed upon ***services are delivered***.

Business Analysis

Purpose: to analyse your business ***needs & recommend solutions***.

Capacity & Performance Management

Purpose: to ensure services ***achieve expected levels***.

Change Enablement (Change Control)

Purpose: to maximise the number of ***successful service & product changes***.

Incident Management

Purpose: to ***minimise negative impact*** of incidents.

IT Asset Management

Purpose: to ***plan & manage*** the life cycle of all IT assets.

Monitoring & Event Management

Purpose: to ***observe*** services & service components.

Problem Management

Purpose: ***reduce incidents*** by identifying ***actual & potential causes***.

Release Management

Purpose: to make new & changed ***services and features available for use***.

Service Catalogue Management

Purpose: to provide a single ***source of consistent information***.

Service Configuration Management

Purpose: to ***ensure accurate & reliable information***.

Service Continuity Management

Purpose: to ***ensure service availability & performance***.

Service Design

Purpose: to ***design products/services fit for purpose & delivery***.

Service Desk

Purpose: to ***capture demand for incident resolution & service requests***.

Service Level Management

Purpose: to ***set business targets*** for service levels.

Service Request Management

Purpose: to ***support the agreed quality of a service***.

Service Validation & Testing

Purpose: to ensure new or changed products & services ***meet defined requirements***.

14 Practices of General Management

Architecture Management

Purpose: explains how ***elements interrelate*** to achieve objectives.

Continual Improvement

Purpose: aligns practices & services with ***changing business needs***.

Information Security Management

Purpose: ***protects information*** needed by an organisation to conduct.

Knowledge Management

Purpose: ***maintains & improves*** information & knowledge.

Measurement & Reporting

Purpose: to support ***decision-making*** and continual improvement.

Organisational Change Management

Purpose: to ensure changes are ***implemented smoothly & successfully***.

Portfolio Management

Purpose: ensures the ***right mix*** of programs, projects, products & services.

Project Management

Purpose: to ensure projects are ***successfully delivered***.

Relationship Management

Purpose: to establish & nurture ***links between organisations and its stakeholders***.

Risk Management

Purpose: to ensure the organisation understands & ***effectively manages risks***.

Strategy Management

Purpose: to ***formulate goals***, course of action & allocate resources.

Service Financial Management

Purpose: to ***support strategies*** & plans for Service Management.

Supplier Management

Purpose: to ensure suppliers & performances are ***managed appropriately***.

Workforce & Talent Management

Purpose: to ensure the ***right people*** have the appropriate ***skills & knowledge***.

Architecture Management

Purpose: explains how ***elements interrelate*** to achieve objectives.

3 Practices of Technical Management

Deployment Management

Purpose: to move ***new or changed*** hardware and software.

Infrastructure & Platform Management

Purpose: to ***oversee & manage*** platforms and infrastructure.

Deployment Management

Purpose: to ensure apps meet the ***needs of the stakeholders***.

Six Activities of the Service Value Chain

Plan: Ensures shared understanding of vision, status, direction & objectives.

Engage: Gathers input from customers to interact with stakeholders to ensure service meets needs.

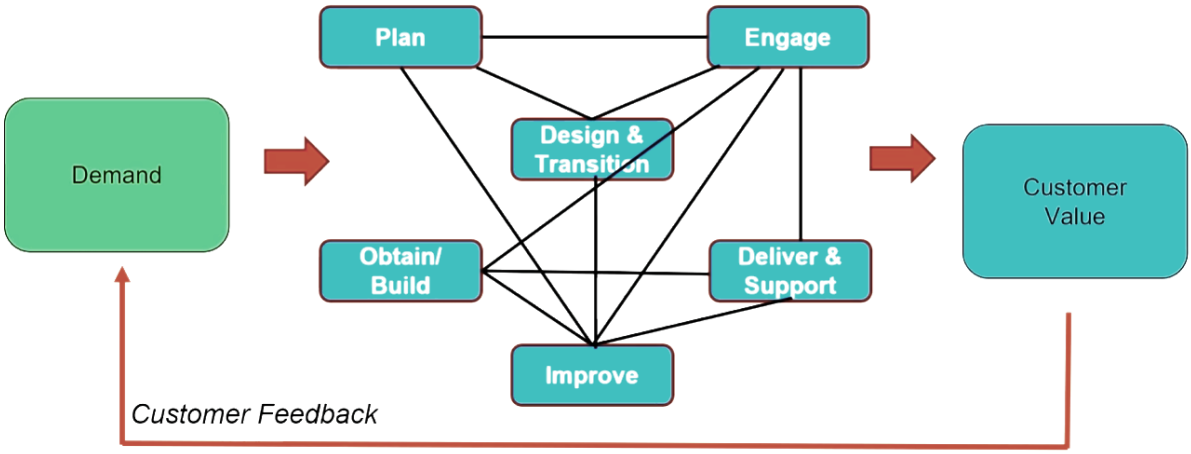
Design & Transition: Develops new or improved IT services while ensuring they meet quality standards.

Obtain/ Build: Acquires or creates necessary components for service delivery.

Deliver & Support: Provides & maintains live IT services for customers & users.

Improve: Continuously enhances services & processes to maintain efficiency and effectiveness.

Interconnected Nature of the Six Values



- SVC is not a linear process.
- An interconnected system.
- Value Streams: sequences of activities.
- Core operating model of ITIL 4.
- Highly dynamic & adaptable.

Different Types of Value Streams

- Value Streams are unique pathways through the Service Value Chain.
- Depends on the type of service being delivered & sequence of activities through the Service Value Chain.
- Service Value Chain provides a structured framework to multiple value streams.

Operational Value Stream

- Delivers ongoing service to customers.
- Continuous service optimisation:
    - Reduces outages.
    - Improves system performance.
    - Enhances customer experience.
    - Analyses service desk metrics, customer feedback & incident reports.

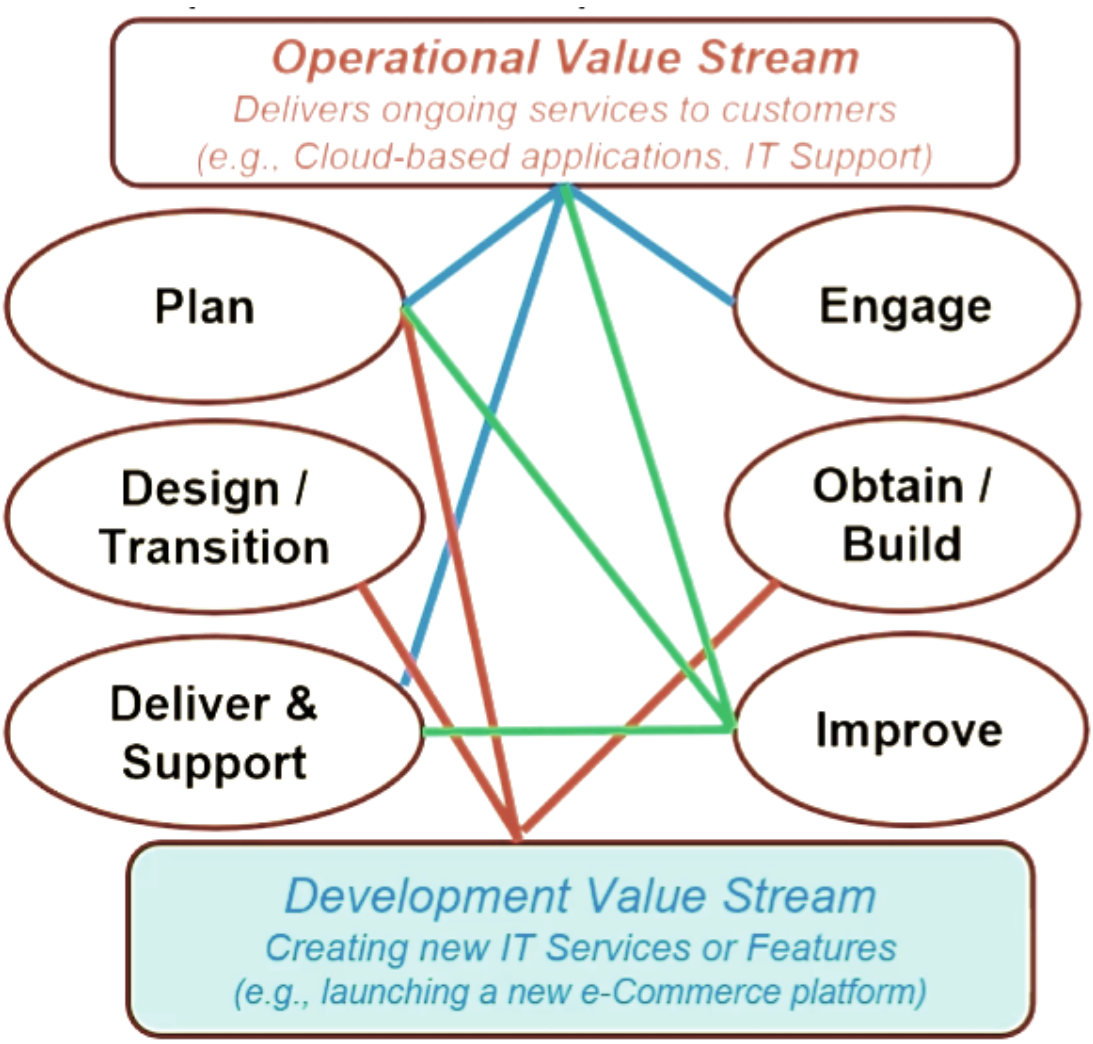
Example: A cloud-based application uses AI-driven monitoring.

Development Value Stream

- Creating new IT services or features where the Service Value Chain dynamically supports both types.
- Ensures a structured but flexible approach to service management.

- For Agile teams to continuously refine software based on market trends.

Example: A new e-commerce platform is improved based on customer behaviour analysis & usability testing.



Purpose of Each Service Value Chain

Plan

Purpose: ensures everyone shared a common understanding.

Key Functions:

- Define strategic objectives.
- Assess current service capabilities.
- Establish governance.
- Forecast demand for services.
- Identify risks & opportunities.

Example: A global e-commerce company plans to support Black Friday sales surge.

Engage

Purpose: ensures continuous communication with customers.

Key Functions:

- Collect business requirement s & user feedback.
- Maintain strong relationships.
- Ensures clear service expectations.
- Foster collaborations between IT & business teams.
- Monitor and analyse customer experience metrics.

Example: A hospital IT department engages with doctors & nurses to understand Electronic Health Record interactions.

Design & Transition

Purpose: focuses on developing new or changed IT services.

Key Functions:

- Develop new services, features or process improvements.
- Ensure compliance.
- Conduct service validation.
- Define clear deployments.
- Minimise service disruption during transition.

Example: A banking institution launches a mobile payment app.

Obtain & Build

Purpose: focuses on acquiring, developing or modifying resources.

Key Functions:

- Develop or purchase software, infrastructure or tools.
- Ensure new components are compatible.
- Leverage automation & DevOps.
- Optimise costs.
- Apply agile & continuous integration.

Example: A telecommunications company decides to build an AI-powered chatbot.

Deliver & Support

Purpose: ensures services are delivered efficiently.

Key Functions:

- Ensure seamless service operations.
- Manage incident resolution.

- Provide technical support.
- Monitor service performance.
- Implement proactive problem management.

Example: A cloud computing provider ensures 24/7 support for enterprise customers.

Improve

Purpose: ensure continuous service enhancement.

Key Functions:

- Conduct post-mortems.
- Implement feedback-driven service improvements.
- Use data-driven decision-making.
- Encourage a culture of continuous learning.
- Align improvements with changing business needs.

Example: An online streaming platform that analyses user engagement metrics.

Purpose of 15 ITIL 4 Practices

Information Security Management

Purpose: To protect IT services, data & systems from security threat and ensure compliance with security regulations.

Key Terminology:

1. CIA Triad
2. Access Control
3. Encryption
4. Security Incident

Scenarios:

1. Enforcement of multi-factor authentication.
2. Implements data encryption.
3. Monitors network activity.

Example: Strengthening IT Security in a Financial Institution to prevent experiencing a data breach due to weak authentication protocols.

Solution:

1. Multi-factor authentication.
2. Security awareness training.
3. Intrusion detection systems.

Relationship Management

Purpose: Establish & maintain strong relationships between IT service providers & stakeholders.

Key Terminology:

1. Stakeholders
2. Service Level Agreements
3. Customer Satisfaction Metrics

Scenarios:

1. IT department holds quarterly meetings with business leaders to ensure service alignment.
2. Software vendor surveys customers.

Example: Retail company faced misalignment between IT services & business goals.

Solution:

1. Created a structured engagement model.
2. Developed a customer feedback loop.

Supplier Management

Purpose: To ensure IT services & products meet business needs.

Key Terminology:

1. Vendor Contracts
2. Service Provider Performance Reviews
3. Third Party Risk Management

Scenarios:

1. Renegotiate cloud service contract.
2. Evaluates software vendors.

Example: Tech startup using multiple cloud service providers but frequent downtime impacts operations.

Solution:

1. Consolidated services to one provider.
2. Introduced performance reviews.

Asset Management

Purpose: Track & manage IT assets.

Key Terminology:

1. Asset Lifecycle Management
2. Configuration Management Database
3. License Compliance

Scenarios:

1. A company tracks laptops & software licenses.
2. An IT team monitors warranty periods.

Example: Manufacturer struggled with untracked IT hardware.

Solution:

1. Implemented an IT asset tracking system.
2. Automated software license renewals.

Release Management

Purpose: To plan, schedule & control new software releases.

Key Terminology:

1. Version Control
2. Release Testing
3. Rollback Plan

Scenarios:

1. A bank deploys security patches.
2. A platform tests a new feature in staging environment.

Example: A company faces bugs in its mobile app updates, frustrating users.

Solution:

1. Implemented a phased rollout strategy.
2. Used comparison testing to validate new features.

Monitoring & Event Management

Purpose: Ensures IT services & infrastructure are continuously observed.

Key Areas:

1. Collect system performance data.
2. Detect & analyse events.
3. Taking appropriate actions.

Key Terminology:

1. Proactive Monitoring (continuously tracking IT services & infrastructure performance)
2. Event Correlation (any occurrence in an IT system that requires attention)
3. Threshold Alerts (limits that trigger an alert when exceeded)

Monitoring:

- **Proactive:** Detect issues before problem.
- **Reactive:** Respond to problems that happen.

Scenarios:

1. A cloud provider monitors CPU usage.
2. A banks IT ream sets up real-time alerts.
3. A retail company monitors website response times.

Example: Healthcare organisation faces frequent system crashes.

Solution:

1. Implemented AI-driven Monitoring & Event Management tools.
2. Set threshold-based alerts.
3. Used event correlation analysis.

Service Configuration Management

Purpose: To ensure accurate & up-to-date information.

Key Terminology:

1. Configuration Items (servers, apps, databases & others)
2. Configuration Management Database
3. Dependency Mapping

Scenarios:

1. IT teams use a CMDB to track infrastructure changes.
2. A telecom company monitors its network components for failures.

Example: A healthcare provider had poor visibility into IT changes.

Solution:

1. Implement a CMDB or tracking system for IT assets.
2. Automated updates for configuration changes.

Deployment Management

Purpose: New or changed IT services deployed into production environments.

Key Terminology:

1. Deployment
2. Release
3. Canary Deployment (release to a small set of users first)
4. Rollback Plan
5. Blue-Green Deployment (software into a test environment)
6. Continuous Deployment
7. Change Windows
8. Deployment Pipeline

Scenarios:

1. Financial institution uses Canary Deployment.
2. E-commerce company uses Blue-Green Deployment.
3. Enterprise IT department automates server patch deployments

Example: A social media company needs to introduce a new AI-powered recommendation engine.

Solution:

1. Release to 5% of users then rollout to everyone else.
2. Have a Rollback Plan.

Continual Improvement

Purpose: To ensure IT services evolve.

Key Terminology:

1. Plan-Do-Check-Act Cycle
2. Service Improvement Plan
3. Key Performance Indicators

Example: IT support tickets took too long to resolve.

Scenarios:

1. Monitoring help desk performance.
2. Improve website performance.

Solution:

1. IT team analysed ticket resolution data.
2. Implementation of AI-chatbots.
3. Introduced weekly review meetings.

Change Enablement (Change Control)

Purpose: Ensure IT changes are made with minimal risk.

Key Terminology:

1. Change Advisory Board
2. Standard Change
3. Emergency Change

Scenarios:

1. A company needs to upgrade its email server.
2. Financial Institution implements a security patch.

Example: Migrating to a new HR system without disruption of payroll.

Solution:

1. Create a rollback plan.

2. Perform testing.
3. Used a phased rollout approach.

Incident Management

Purpose: restore service operations as soon as possible.

Key Terminology:

1. Incident Ticketing System
2. Escalation Matrix
3. Major Incident

Scenarios:

1. A company website crashes during heavy load.
2. An employee's laptop stops working.

Example: Manufacturing company experiences a network outage.

Solution:

1. Incident response team identify root cause.
2. Implemented a temporary workaround.
3. Replace faulty hardware.

Problem Management

Purpose: Identify root cause of recurring IT issues.

Key Terminology:

1. Root Cause Analysis
2. Known Error Database
3. Proactive
4. Reactive

Scenarios:

1. Repeated email outages.
2. IT discovers a bug in accounting system.

Example: VPN connectivity issues impacting remote work.

Solution:

1. Conducted RCS & found outdated firmware.
2. Upgraded servers & documented best practices in the Known Error Database.
3. Provide VPN training to employees.

Service Request Management

Purpose: To handle routine IT service requests such as password resets, software installations & access requests.

Key Terminology:

1. Self-Service Portal
2. Service Catalogue
3. Automated Workflows

Scenarios:

1. Employees request access to shared drives.
2. A company automates password resets.

Example: A university IT department overwhelmed with password reset requests.

Solution:

1. Implemented a self-service password reset tool.
2. Created a knowledgebase with troubleshooting guides.

Service Desk

Purpose: To act as the single point of contact for IT issues to help users resolve issues.

Key Terminology:

1. Tiered Support Model
2. Incident
3. Service Request

Scenarios:

1. A service desk agents help troubleshoot Wi-Fi issues for employees.
2. Customer calls the IT help desk to unlock their account.

Example: A multinational company needs 24/7 IT support for employees.

Solution:

1. Implement AI chatbots for common issues.
2. Established regional support teams for follow-ups.

Service Level Management

Purpose: To define & manage Service Level Agreements to ensure IT services meet performance expectations.

Key Terminology:

1. Service Level Agreement
2. Operational Level Agreements

Scenarios:

1. A cloud provider guarantees 99.9% uptime in its SLAs.
2. A company monitoring service deployments to check for uptime.

Example: Retail company notices IT service provider didn't meet agreed response times.

Solution:

1. Adjust SLA to include faster response times.
2. Implemented penalties for SLA breaches.